

Material response: From classical to quantum regime

Week

II

During the last week, we introduced fundamental concepts of optics to establish a useful landscape for our curricular unit and the key problems we will tackle during the semester. As said, the first part of our course will focus on introducing a quantum formalism to describe matter when interacting with light described in a classical manner. In practice, this **semiclassical approach** starts from the Maxwell wave equation to describe the propagation of the electromagnetic field, and aims to derive a model for the material polarization $\mathbf{P}$ that acts as a source term for the wave equation and accounts for light-matter interaction.

At first sight the answer seems straightforward. For example, for a linear dielectric we may simply write $\mathbf{P} = \epsilon_0 \chi \mathbf{E}$ and proceed with the analysis. Indeed, this already explains much of ordinary optics around us via empirical matching of the parameter χ . However, susceptibility χ can also vary dramatically over a very narrow frequency interval in the presence of a resonance (which indeed occurs a lot at frequencies in the optical part of the electromagnetic spectrum), and its real and imaginary parts determine quite different physical effects.

This week, we will tackle this problem by first introducing a classical model - the Lorentz oscillator - and proceed to discuss its limitations and set the stage for quantum mechanics. We will then introduce a basic quantum formalism and connect the material polarization with the expectation value of a microscopic dipole, allowing to establish a first important pillar for our semiclassical description of atom-field interaction.

1 Material polarization and optical response

The wave equation derived in the first week

$$\nabla^2 \mathbf{E} - \frac{1}{c^2} \frac{\partial^2 \mathbf{E}}{\partial t^2} = \mu_0 \frac{\partial^2 \mathbf{P}}{\partial t^2}. \quad (1.1)$$

will be our starting point for our analysis. As already discussed, in this form, the polarization appears explicitly as a source of the electromagnetic

field. If it is not known, Maxwell's equations alone are insufficient, but if $\mathbf{P}$ is known, the propagation problem is in principle closed. For a generic local and instantaneous electric response, the material polarization may be expanded in powers of the electric field,

$$P_i = \epsilon_0 \left[\sum_j \chi_{ij}^{(1)} E_j + \sum_{jk} \chi_{ijk}^{(2)} E_j E_k + \sum_{jkl} \chi_{ijkl}^{(3)} E_j E_k E_l + \cdots \right]. \quad (1.2)$$

being tensors $\chi^{(n)}$ the n th-order electric susceptibilities. In general, the response does not need to be isotropic, local, or instantaneous, which can make the problem even more complex. For example, already at first order, this results into a complex formalism, e.g.

$$P_i(\mathbf{r}, t) = \epsilon_0 \sum_j \int d^3 r' \int_{-\infty}^t dt' \chi_{ij}^{(1)}(\mathbf{r}, \mathbf{r}'; t - t') E_j(\mathbf{r}', t'), \quad (1.3)$$

where causality has been made explicit by restricting the response to earlier times $t' \leq t$. Spatial nonlocality means that the response at $\mathbf{r}$ may depend on the field at other positions $\mathbf{r}'$, while temporal nonlocality means that the material retains memory of the field at earlier times¹. Fortunately, for most of the effects we usually deal with, these kind of effects can be safely disregarded. We will start then our analysis by considering a **homogeneous, isotropic, linear medium**:

$$\mathbf{P}(\omega) = \epsilon_0 \chi(\omega) \mathbf{E}(\omega), \quad (1.4)$$

where the susceptibility is allowed to depend on the frequency of light to account for possible resonant effects.

¹ for example in photorefractive crystals, or when optical pulses are comparable with population dynamics timescales

Definition. Linear Electric susceptibility: In a linear medium, the electric susceptibility $\chi(\omega)$ is the response function relating the Fourier component of the induced polarization to the corresponding Fourier component of the applied electric field. For an isotropic medium χ is a scalar, whereas for an anisotropic medium it is a rank-2 tensor χ_{ij} . Besides, the electric susceptibility function is, in general, complex,

$$\chi(\omega) = \chi'(\omega) + i\chi''(\omega). \quad (1.5)$$

with χ' relating to the phase propagation and **dispersion**, whereas χ'' describes irreversible transfer of energy between field and matter, appearing as **absorption or gain**. We shall refer also that these two terms are not independent due to causality. Indeed these are related through the

Kramers–Kronig relations,

$$\chi'(\omega) = \frac{2}{\pi} \mathcal{P} \int_0^\infty \frac{\Omega \chi''(\Omega)}{\Omega^2 - \omega^2} d\Omega \quad (1.6)$$

$$\chi''(\omega) = -\frac{2\omega}{\pi} \mathcal{P} \int_0^\infty \frac{\chi'(\Omega)}{\Omega^2 - \omega^2} d\Omega \quad (1.7)$$

whose derivation and further discussion we will leave out of the scope of this curricular unit.

2 Dispersion and absorption

To analyze the effect of each susceptibility term, let's return to the electromagnetic field propagation. Considering a light beam propagating along z and linearly polarized along the x direction, with $E_0(z)$ being a slowly varying envelope on the scale k^{-1} (disregarding the transverse direction for now), we have

$$\mathbf{E}(x, y, z, t) = \frac{1}{2} \hat{\mathbf{x}} E_0(z) e^{i(kz - \omega t)} + \text{c.c.} \quad (2.1)$$

which results into a polarization term

$$\mathbf{P}(z, t) = \frac{1}{2} \hat{\mathbf{x}} \epsilon_0 \chi(\omega) E_0(z) e^{i(kz - \omega t)} + \text{c.c.} \quad (2.2)$$

Substitution into the wave equation,

$$\frac{\partial^2 E}{\partial z^2} - \frac{1}{c^2} \frac{\partial^2 E}{\partial t^2} = \mu_0 \frac{\partial^2 P}{\partial t^2}, \quad (2.3)$$

gives an exact equation for the envelope,

$$\frac{d^2 E_0}{dz^2} + 2ik \frac{dE_0}{dz} - k^2 E_0 + \frac{\omega^2}{c^2} E_0 = -\frac{\omega^2}{c^2} \chi(\omega) E_0. \quad (2.4)$$

which, applying the slowly varying envelope approximation², results into $2 \left| \frac{d^2 E_0}{dz^2} \right| \ll \left| 2ik \frac{dE_0}{dz} \right|$

$$2ik \frac{dE_0}{dz} - \left(k^2 - \frac{\omega^2}{c^2} \right) E_0 = -\frac{\omega^2}{c^2} \chi(\omega) E_0. \quad (2.5)$$

We now proceed by analyzing the role that each part (real *vs* imaginary) of the susceptibility plays.

2.1 Real part of the susceptibility and dispersion

The real part of Eq. 2.5 determines the relation between k and ω (also called dispersion relation). For a weakly absorbing medium we may consider k to

be a real number which shall satisfy

$$k^2 = \frac{\omega^2}{c^2} [1 + \chi'(\omega)]. \quad (2.6)$$

Introducing the refractive index,

$$k(\omega) = n(\omega) \frac{\omega}{c}, \quad (2.7)$$

we have, under the same weak-loss approximation,

$$n(\omega) = \sqrt{1 + \chi'(\omega)}, \quad (2.8)$$

introducing the concept of dispersion *dispersion*.

Definition. Dispersion: A medium is dispersive when the wavenumber is not a linear function of frequency, or equivalently when $n = n(\omega)$, i.e. the index of refraction depends on the frequency. This means that different spectral components then accumulate phase at different rates during propagation.

The refractive index also determines the velocity at which the phase propagates. Indeed, looking at the exponential part of the monochromatic ansatz, i.e. $e^{i(kz - \omega t)}$, we can also see that the wavefront (i.e. surface with same phase³) moves along z with a **phase velocity** given by

³ $kz - \omega t = \text{constant}$

$$v_p(\omega) = \frac{\omega}{k} = \frac{c}{n(\omega)}. \quad (2.9)$$

But what happens to a wavepacket, e.g. what determines the propagation velocity of the maximum? To explore this we first see that a general one-dimensional wave packet may be written as a superposition of plane waves:

$$E(z, t) = \int A(k) e^{i(kz - \omega(k)t)} dk. \quad (2.10)$$

Specifically, when the spectrum $A(k)$ is narrowly centered around k_0 ⁴, the dispersion relation can be expanded to first order:⁵

⁴ e.g. a short optical pulse

⁵ For ultrashort pulses, we shall expand further

$$\omega(k) \approx \omega_0 + \left. \frac{d\omega}{dk} \right|_{k_0} (k - k_0). \quad (2.11)$$

$$k(\omega) = k_0 + k_1(\omega - \omega_0) + \frac{1}{2}k_2(\omega - \omega_0)^2 + \dots,$$

Substituting this approximation yields

$$E(z, t) \approx e^{i(k_0 z - \omega_0 t)} F(z - v_g t), \quad (2.12)$$

with k_1 being the group velocity and k_2 as a group-velocity dispersion parameter, responsible, at lowest order, for temporal reshaping and broadening of a pulse.

where

$$v_g = \left. \frac{d\omega}{dk} \right|_{k_0} = \frac{c}{n(\omega_0) + \omega_0 \left. \frac{dn}{d\omega} \right|_{\omega_0}}. \quad (2.13)$$

is called the *group velocity*, meaning that the envelope propagates with a distinct velocity from the rapidly oscillating carrier (which still propagates with the phase velocity). Note that in a nondispersive medium, where $\omega = (c/n)k$, one still finds $v_p = v_g = \frac{c}{n}$. Also, if the susceptibility changes abruptly (e.g. in the presence of a resonance), this means that the group velocity can also change in a dramatic manner, which will be something we will explore in the context of this curricular unit later.

2.2 Imaginary susceptibility and absorption

Looking now at the imaginary part of Eq. 2.5, it is easy to relate it to the variation of the field envelope amplitude. Choosing k according to the real dispersion relation, one obtains

$$\frac{dE_0}{dz} = -\frac{\omega^2}{2kc^2} \chi''(\omega) E_0. \quad (2.14)$$

In terms of the optical intensity, which is proportional to $|E_0|^2$, we get

$$\frac{dI}{dz} = -\alpha(\omega) I, \quad (2.15)$$

where

$$\alpha(\omega) = \frac{\omega^2}{kc^2} \chi''(\omega) \quad (2.16)$$

is the absorption coefficient in this approximation. The solution of this differential equation corresponds to the Beer-Lambert law

$$I(z) = I(0)e^{-\alpha z}. \quad (2.17)$$

Thus, for the $e^{-i\omega t}$ convention used here, $\chi'' > 0$ corresponds to **absorption** and $\chi'' < 0$ to **gain**. Note that an equivalent description may use a complex refractive index $\tilde{n}(\omega) = n(\omega) + i\kappa(\omega)$ which is associated with an $\alpha = \frac{2\omega\kappa}{c}$. For weak absorption, $\tilde{n}^2 = 1 + \chi$, this gives a relation with the susceptibility term as $2n\kappa \simeq \chi''$.

3 Classical model for resonant response: the Lorentz oscillator

So far, the susceptibility has been introduced as a macroscopic response function: we know what its real and imaginary parts do, but we have not yet constructed a model that predicts its frequency dependence. Looking at the problem from a classical perspective, the simplest model is to treat matter response as the response of a bound electron as a driven and damped harmonic oscillator.

Let's start by considering an electron of charge $-e$ and mass m bound to a much heavier positive center that is forced by an applied electric field (e.g. a light wave). Taking the positive center to remain fixed, modelling the restoring force as linear in the displacement x (natural frequency ω_0), and including a viscous damping force term (γ) to model dissipation, we obtain a typical equation of motion for a driven damped harmonic oscillator,

$$m\ddot{x} + m\gamma\dot{x} + m\omega_0^2x = -eE_0e^{-i\omega t}. \quad (3.1)$$

Tackling this equation in the Fourier space, $x(t) \rightarrow X(\omega)$, it is trivial to obtain by direct substitution into Eq. 3.1 the relation

$$(\omega_0^2 - \omega^2 - i\gamma\omega) X(\omega) = -\frac{e}{m}E_0 \Rightarrow X(\omega) = -\frac{eE_0/m}{\omega_0^2 - \omega^2 - i\gamma\omega}, \quad (3.2)$$

meaning that the displacement amplitude depends on the wave frequency ω . Furthermore, defining the microscopic electric dipole moment of this system as the displacement from the center and going to the frequency domain, we obtain

$$p(t) = -ex(t) \Rightarrow p(\omega) = -eX(\omega) = \frac{e^2/m}{\omega_0^2 - \omega^2 - i\gamma\omega}E_0. \quad (3.3)$$

If the material contains a number density $\mathcal{N}$ of identical, independent oscillators, the macroscopic polarization is $P(\omega) = \mathcal{N}p(\omega)$, which comparing with the definition of the linear polarization, $P = \epsilon_0\chi(\omega)E_0$ giving:

Lorentz-oscillator susceptibility:

$$\chi(\omega) = \frac{\mathcal{N}e^2}{\epsilon_0 m} \frac{1}{\omega_0^2 - \omega^2 - i\gamma\omega} \quad (3.4)$$

for a single species of identical classical bound oscillators.

3.1 Real and imaginary parts of the susceptibility

Writing

$$\chi(\omega) = \chi'(\omega) + i\chi''(\omega), \quad (3.5)$$

it is trivial to obtain

$$\chi'(\omega) = C \frac{\omega_0^2 - \omega^2}{(\omega_0^2 - \omega^2)^2 + \gamma^2 \omega^2}, \quad (3.6)$$

and

$$\chi''(\omega) = C \frac{\gamma \omega}{(\omega_0^2 - \omega^2)^2 + \gamma^2 \omega^2}. \quad (3.7)$$

with $C \equiv \frac{\mathcal{N}e^2}{\epsilon_0 m}$. Looking at the formula (and further assisted by Figure [ref]) it is easy to see that the absorptive part χ'' peaks around ω_0 , while the dispersive part χ' changes rapidly and changes sign across the resonance.

Defining a detuning $\Delta \equiv \omega - \omega_0$ ($|\Delta| \ll \omega_0$, $\gamma \ll \omega_0$). to further describe the behavior near the resonance⁶ Eq. 3.4 becomes approximately

$$\omega_0^2 - \omega^2 \simeq -2\omega_0 \Delta$$

$$\chi(\omega) \simeq -\frac{\mathcal{N}e^2}{2\epsilon_0 m \omega_0} \frac{1}{\Delta + i\gamma/2}. \quad (3.8)$$

Consequently,

$$\chi''(\Delta) \propto \frac{\gamma/2}{\Delta^2 + (\gamma/2)^2}, \quad (3.9)$$

which is a Lorentzian absorption line with full width at half maximum γ in this simple model. The corresponding real part is antisymmetric in detuning,

$$\chi'(\Delta) \propto -\frac{\Delta}{\Delta^2 + (\gamma/2)^2}, \quad (3.10)$$

which means that the induced dipole may be either in-phase or π out-of-phase depending on the signal of the detuning. Besides, on resonance, the real part of the susceptibility vanishes $\chi'(\omega_0) = 0$ while $\chi''(\omega_0) = \frac{\mathcal{N}e^2}{\epsilon_0 m \gamma \omega_0}$. The induced dipole is then $\pi/2$ out of phase with the driving field, a phase lag that is a classical signature of maximum energy transfer from the field to the damped oscillator.

Multiple Resonances: Real materials may contain many possible resonances. A classical generalization is to sum oscillator contributions,

$$\chi(\omega) = \frac{\mathcal{N}e^2}{\epsilon_0 m} \sum_j \frac{f_j}{\omega_j^2 - \omega^2 - i\gamma_j \omega}, \quad (3.11)$$

where the coefficients f_j , γ_j are characteristics for each oscillator contri-

bution.

3.2 Where the classical model fails

Overall, the Lorentz model gives a qualitatively correct picture of simple resonant absorption and dispersion and remains a standard and very practical way of parameterizing dielectric response in such conditions. Actually, the model fits the absorption lineshape quite well for simple systems, as illustrated in the example in Figure [ref]. However, it still has some limitations:

- The parameters (restoring frequency ω_0 and damping rate γ) need to be inserted by hand and **are not predicted by classical mechanics**. Indeed, classical mechanics does not predict the discrete set of internal frequencies observed in atoms and molecules, nor provides a complete microscopic theory of radiative and non-radiative relaxation.
- The model **does not explain atomic stability**, as a classical electron orbiting a nucleus is not a stationary state, and eventually, in the presence of damping and in the absence of a forcing field, collapses to the center.
- The mechanical spring picture **does not predict a varying response for different optical transitions**. In particular, the classical oscillator does not provide the symmetry selection rules that determine when an electric-dipole transition is forbidden.
- The model fails to account for population **saturation and nonlinear optical effects under strong field excitation**. Because a classical harmonic oscillator's response scales linearly with the driving field amplitude, the Lorentz model cannot predict power broadening, multiphoton processes, or the intensity-dependent bleaching of absorption.
- The Lorentz model also **cannot accurately capture more complex spectral lineshapes or interactions**. Asymmetrical profiles arising from quantum interference (such as Fano resonances), inhomogeneous Doppler broadening, multi-level cascade dynamics, and non-Markovian relaxation mechanisms lie strictly beyond the scope of a single linear classical oscillator.

These shortcomings point toward the need for a more appropriate microscopic description. From quantum mechanics, we already know that the electron occupies quantum states with definite energies, and we may think that the optical field should couple these states through matrix elements of physical observables. This sets the stage for the semiclassical approach of

quantum optics: the optical field will remain classical, while the material degrees of freedom will be described by quantum mechanics.

4 Overview of Quantum Mechanics

4.1 Matter waves and stationary states

Quantum mechanics provides a fundamental framework for describing matter at the microscopic scale, where wave-particle duality plays a central role. In this regime, a particle moving with momentum p is intrinsically associated with a de Broglie wavelength:

$$\lambda_B = \frac{h}{p}. \quad (4.1)$$

This duality introduces a cornerstone concept of quantum theory: within a bound system, only wave patterns that satisfy the system's boundary conditions can form stationary states. Mathematically, these allowed physical states will correspond to the eigenfunctions of the Hamiltonian operator, which is itself related with the energy of the system as we will see shortly. As we are expecting states, we introduce a framework to represent quantum states as a vector ket $|\psi(t)\rangle$. In the position representation, it is related with the wavefunction as

$$\psi(\mathbf{r}, t) \equiv \langle \mathbf{r} | \psi(t) \rangle. \quad (4.2)$$

The wavefunction is a complex probability amplitude, and the probability density for finding the particle near $\mathbf{r}$ at time t is obtained by computing the local density

$$\rho_p(\mathbf{r}, t) = |\psi(\mathbf{r}, t)|^2. \quad (4.3)$$

This convention imposes the normalization

$$\int_{\mathbb{R}^3} |\psi(\mathbf{r}, t)|^2 d^3r = 1, \quad (4.4)$$

i.e. the probability of finding the particle in all the space considered should be one.

4.2 Time evolution and energy eigenstates

A central goal of quantum mechanics is to determine the state vector of a physical system and predict its time evolution. Historically, Erwin Schrödinger formulated his wave equation in 1925 by building on the Louis de Broglie's matter-wave hypothesis ($E = \hbar\omega$ and $p = \hbar k$). By considering a monochro-

matic plane wave, he extended it to matter-wave interpretation,

$$Ae^{i(kx-\omega t)} \rightarrow Ae^{\frac{i}{\hbar}(px-Et)} = \psi(x, t). \quad (4.5)$$

Then, applying non-relativistic energy conservation ($E = \frac{p^2}{2m} + V$) and observing that $E = i\hbar \frac{\partial}{\partial t} \psi(x, t)$, Schrödinger derived a dynamic equation for matter waves in the form of what we call time-dependent Schrödinger equation:

$$i\hbar \frac{\partial}{\partial t} |\psi(t)\rangle = \hat{H} |\psi(t)\rangle. \quad (4.6)$$

Definition. Stationary States: For a time-independent Hamiltonian $\hat{H}$, the stationary states $|n\rangle$ are eigenstates of the Hamiltonian satisfying the time-independent Schrödinger equation:

$$\hat{H} |n\rangle = E_n |n\rangle, \quad (4.7)$$

associated with a fixed energy E_n which implies that their time evolution is purely harmonic:

$$|n, t\rangle = e^{-iE_n t/\hbar} |n\rangle. \quad (4.8)$$

Note that because the energy eigenstates form a complete orthonormal basis,

$$\langle m | n \rangle = \delta_{mn}, \quad \sum_n |n\rangle \langle n| = \hat{I}, \quad (4.9)$$

An arbitrary pure state can also be expanded as a superposition of these states:

$$|\psi(t)\rangle = \sum_n c_n e^{-iE_n t/\hbar} |n\rangle, \quad (4.10)$$

with the normalization condition:

$$\sum_n |c_n|^2 = 1. \quad (4.11)$$

Upon a measurement of energy, $|c_n|^2$ represents the probability of obtaining the energy eigenvalue E_n . In particular, it is important to emphasize the physical distinction between a single stationary state and a superposition of stationary states associated with its time dynamics. A single stationary state exhibits a time-independent spatial probability density, whereas a coherent superposition contains relative phase terms that evolve over time at the Bohr transition frequencies:

$$\omega_{nm} = \frac{E_n - E_m}{\hbar}. \quad (4.12)$$

These dynamically evolving relative phases are precisely the microscopic ori-

gin of an oscillating electric dipole moment in atomic and molecular systems.

4.3 Quantum observables

In the quantum formalism, any measurable physical quantity is represented by a Hermitian operator $\hat{O}$ that is applied to the quantum state. By definition, the eigenstates of the operator satisfy

$$\hat{O}|o_n\rangle = o_n|o_n\rangle, \quad (4.13)$$

where the eigenvalues o_n **must be real⁷ and correspond to possible measurement outcomes**. If the system is in a normalized state $|\psi\rangle$, we define the mean expectation value of $\hat{O}$,

⁷ due to the fact that the operator is hermitic, $\hat{O} = \hat{O}^\dagger$

Expectation value of an observable:

$$\langle \hat{O} \rangle = \langle \psi | \hat{O} | \psi \rangle \quad (4.14)$$

or, in the position representation,

$$\langle \hat{O} \rangle = \int d^3r \psi^*(\mathbf{r}, t) \hat{O} \psi(\mathbf{r}, t). \quad (4.15)$$

It is easy to see that if the state is expanded in eigenstates of the observable,

$$|\psi\rangle = \sum_n a_n |o_n\rangle \Rightarrow \langle \hat{O} \rangle = \sum_n |a_n|^2 o_n, \quad (4.16)$$

which means that the **expectation value is not the result of a single measurement but corresponds to an ensemble mean obtained from repeated measurements on identically prepared systems**.

For what matters for the semiclassical approach of quantum optics, the observable of greatest importance will be the electric dipole moment.

5 Material polarization as a quantum observable

5.1 Microscopic dipole operator

For a single electron of charge $-e$ measured relative to a chosen origin, the electric-dipole operator $\hat{\mathbf{p}}$ is defined by promoting the distance $\mathbf{r}$ of the typical classical dipole formula to an operator

$$\mathbf{p} = -e\mathbf{r} \Rightarrow \hat{\mathbf{p}} = -e\hat{\mathbf{r}}. \quad (5.1)$$

The expectation value of this operator in the state $|\psi(t)\rangle$ is thus

$$\langle \hat{\mathbf{p}} \rangle = -e \langle \psi(t) | \hat{\mathbf{r}} | \psi(t) \rangle, \quad (5.2)$$

or equivalently

$$\langle \hat{\mathbf{p}} \rangle = -e \int d^3r \psi^*(\mathbf{r}, t) \mathbf{r} \psi(\mathbf{r}, t). \quad (5.3)$$

For many-electron systems the total electric dipole is obtained by summing the charge-weighted position operators of all particles, together with the nuclear contribution when required. If the material contains a number density $\mathcal{N}$ of identical, independently responding microscopic systems, the macroscopic polarization can then be defined as:

Expectation value of the material polarization:

$$\boxed{\mathbf{P}(\mathbf{r}, t) = \mathcal{N} \langle \hat{\mathbf{p}}(t) \rangle} \quad (5.4)$$

To understand the role of material polarization as a source for a given frequency, we now look at what happens for a general superposition state. Considering the time dynamics,

$$|\psi(t)\rangle = \sum_n c_n e^{-i\omega_n t} |n\rangle, \quad \omega_n = \frac{E_n}{\hbar}, \quad (5.5)$$

it is straightforward to obtain

$$\langle \hat{\mathbf{p}}(t) \rangle = \sum_{m,n} c_m^* c_n e^{i\omega_m t} e^{-i\omega_n t} \langle m | \hat{\mathbf{p}} | n \rangle \quad (5.6)$$

$$= \sum_{m,n} c_m^* c_n \mathbf{p}_{mn} e^{-i\omega_{nm} t}, \quad (5.7)$$

where we defined a **dipole matrix element**,

$$\mathbf{p}_{mn} \equiv \langle m | \hat{\mathbf{p}} | n \rangle \quad (5.8)$$

which oscillate with transition frequencies

$$\omega_{nm} = \omega_n - \omega_m = \frac{E_n - E_m}{\hbar}. \quad (5.9)$$

that correspond to the Bohr frequencies and may belong to the light (UV-VIS-IR) part of the electromagnetic spectrum, acting thus a source in the Maxwell's wave equation. Note also that the diagonal terms $\mathbf{p}_{nn}$, represent permanent dipole moments of stationary states, whereas the off-diagonal terms $\mathbf{p}_{mn}, m \neq n$, are transition dipole moments. When two states are coherently present, the off-diagonal terms oscillate at the Bohr frequency ω_{nm} .

5.2 Symmetry, parity, and electric-dipole selection rules

Calculating an atomic wavefunction exactly is often mathematically challenging and, for multi-electron or complex systems, analytically impossible. Before attempting an exact calculation, it is essential to simplify the problem by analyzing the natural symmetries of the system. While a complete treatment of physical symmetries relies on group theory, much can be learned by considering specific discrete symmetries directly.

In particular, spatial inversion symmetry plays a fundamental role in atomic and molecular systems. To explore spatial inversion, we introduce the parity operator $\hat{\Pi}$, which acts on a wavefunction by inverting all spatial coordinates ($\mathbf{r} \rightarrow -\mathbf{r}$):

$$\hat{\Pi}\psi(\mathbf{r}) = \psi(-\mathbf{r}). \quad (5.10)$$

It is easy to see that applying parity twice returns the original state, and thus the eigenvalues can only be

$$\hat{\Pi}^2 = \hat{I} \Rightarrow \pi = \pm 1. \quad (5.11)$$

A parity eigenstate therefore satisfies

$$\psi(-\mathbf{r}) = +\psi(\mathbf{r}), \quad \text{even parity} \quad (5.12)$$

$$\psi(-\mathbf{r}) = -\psi(\mathbf{r}), \quad \text{odd parity.} \quad (5.13)$$

Also, if the Hamiltonian is invariant under inversion, we have

$$\hat{H}(-\mathbf{r}) = \hat{H}(\mathbf{r}) \Rightarrow [\hat{H}, \hat{\Pi}] = 0. \quad (5.14)$$

In this case, the energy eigenstates may, therefore, be divided in two groups of definite parity. For a non-degenerate energy eigenstate, this follows immediately; in a degenerate subspace, one may choose a basis that simultaneously diagonalizes $\hat{H}$ and $\hat{\Pi}$.

5.3 Parity of the electric dipole operator and selection rules

The position operator is odd under inversion, and therefore the electric-dipole operator is also odd, i.e.

$$\hat{\Pi}\hat{\mathbf{r}}\hat{\Pi}^{-1} = -\hat{\mathbf{r}} \Rightarrow \hat{\Pi}\hat{\mathbf{p}}\hat{\Pi}^{-1} = -\hat{\mathbf{p}}. \quad (5.15)$$

This simple transformation property has two immediate consequences:

A parity eigenstate has no permanent electric dipole: let $|n\rangle$ have definite parity π_n . Then

$$\mathbf{p}_{nn} = \langle n | \hat{\mathbf{p}} | n \rangle \quad (5.16)$$

$$= \langle n | \hat{\Pi}^{-1} (\hat{\Pi} \hat{\mathbf{p}} \hat{\Pi}^{-1}) \hat{\Pi} | n \rangle \quad (5.17)$$

$$= \pi_n^2 \langle n | (-\hat{\mathbf{p}}) | n \rangle \quad (5.18)$$

$$= -\mathbf{p}_{nn}. \quad (5.19)$$

Hence

$$\boxed{\mathbf{p}_{nn} = 0}, \quad (5.20)$$

which means that an **inversion-symmetric system in a stationary parity eigenstate has no permanent electric dipole moment.**

Electric-dipole transitions connect states of opposite parity in a system with inversion symmetry: Consider a transition matrix element

$$\mathbf{p}_{mn} = \langle m | \hat{\mathbf{p}} | n \rangle. \quad (5.21)$$

If $|m\rangle$ and $|n\rangle$ have parity eigenvalues π_m and π_n , then

$$\mathbf{p}_{mn} = \langle m | \hat{\Pi}^{-1} (\hat{\Pi} \hat{\mathbf{p}} \hat{\Pi}^{-1}) \hat{\Pi} | n \rangle \quad (5.22)$$

$$= -\pi_m \pi_n \mathbf{p}_{mn}. \quad (5.23)$$

A non-zero matrix element therefore requires

$$-\pi_m \pi_n = 1, \quad (5.24)$$

or

$$\boxed{\pi_m \pi_n = -1}. \quad (5.25)$$

Thus electric-dipole transitions connect eigenstates of opposite parity.

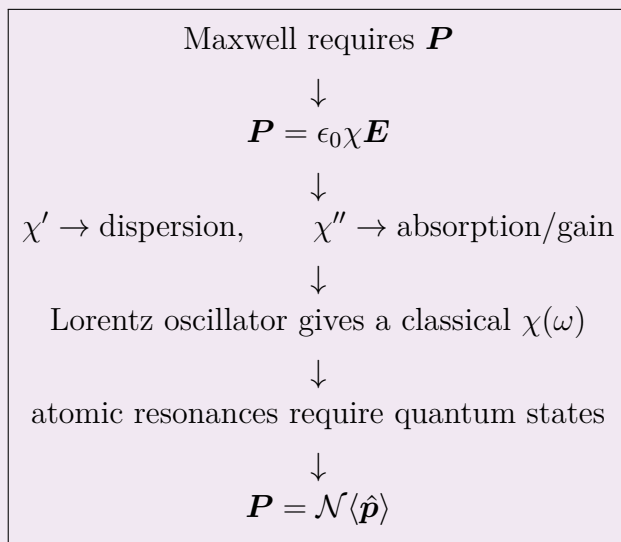
Example 1. Hydrogenic parity. Hydrogen has inversion symmetry and atomic orbitals can be shown to have parity $(-1)^\ell$. Therefore an electric-dipole transition requires a change from even to odd parity or vice versa. The familiar rule $\Delta\ell = \pm 1$ is consistent with this parity requirement. Thus a $1s \rightarrow 2p$ transition is electric-dipole allowed, while a $1s \rightarrow 2s$ transition is forbidden at electric-dipole order. However, the latter may still occur through weaker higher-order processes.

Note that parity is only one part of the full set of optical selection rules.

Rotational symmetry and conservation of angular momentum provide additional restrictions, including rules for magnetic quantum numbers and the dependence on the polarization of the driving field. Those ideas will be introduced when we need them during the semester. But one important note at this point is that the magnitude of the optical response is controlled not only by an energy resonance but also by whether the relevant dipole matrix element is permitted by symmetry or not, which was not considered in the classical counterpart.

6 Concluding remarks

Learning Objectives 1. The main chain of ideas developed this week is



After completing this week, you should be able to

- formulate the optical propagation problem in terms of the material polarization and the electric susceptibility;
- relate the real and imaginary parts of $\chi(\omega)$ to dispersion and absorption or gain;
- derive the frequency-dependent susceptibility of the classical Lorentz oscillator and identify its resonant behaviour;
- state precisely which aspects of atomic optical response are not explained by the classical oscillator model;
- introduce a quantum state through its wavefunction and compute expectation values of observables;

- identify the electric dipole as the microscopic observable underlying the macroscopic material polarization;
- understand the relation between inversion symmetry and parity selection rules for electric-dipole transitions.

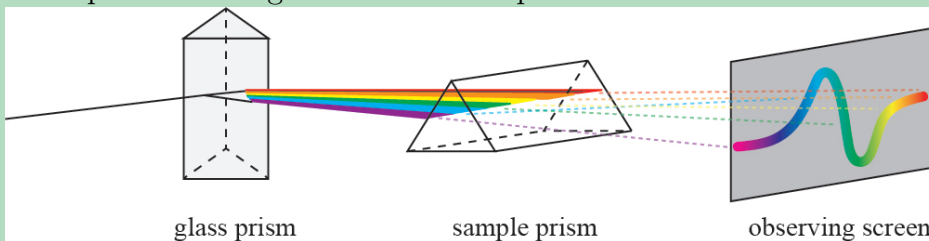
Additional practice exercises

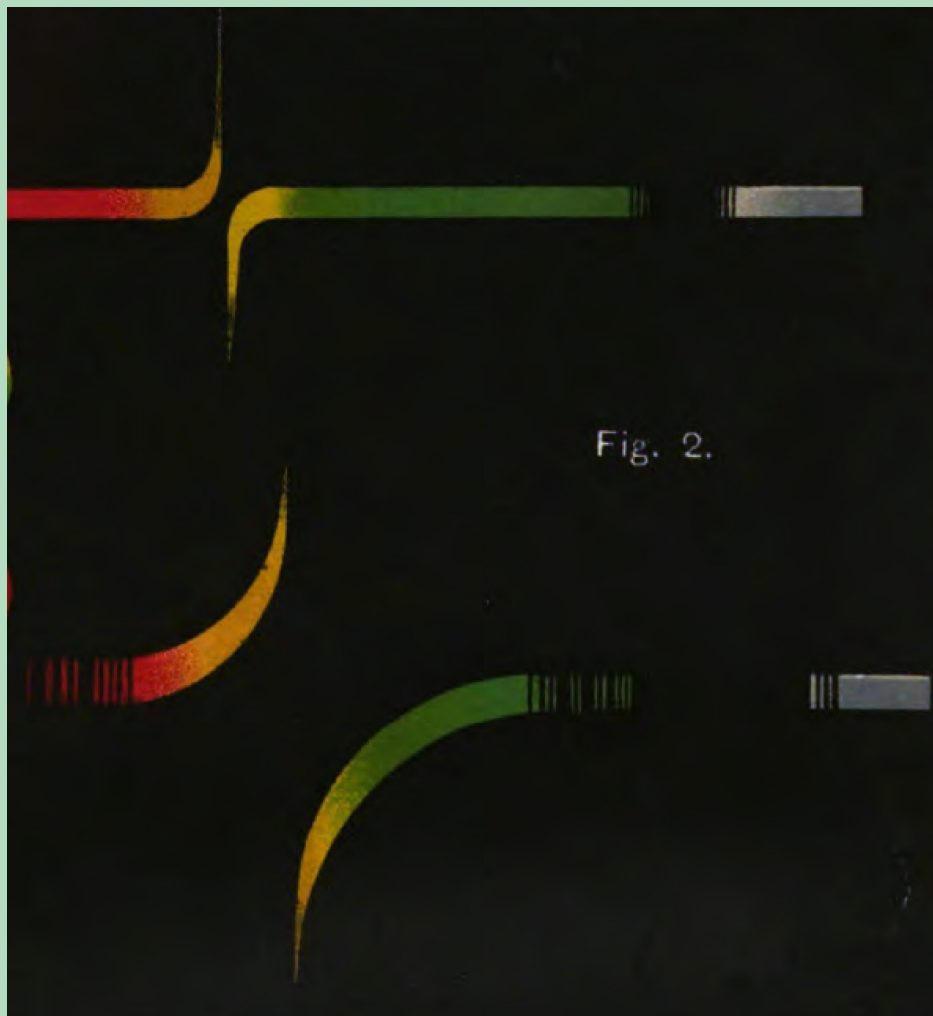
Exercise 1. Starting from the source-driven wave equation and the ansatz $E(z, t) = E_0(z)e^{i(kz - \omega t)}$, derive Eq. 2.4. State explicitly which term is neglected under the slowly varying envelope approximation and formulate the approximation in terms of a characteristic envelope length L_z .

Exercise 2. For a complex susceptibility $\chi = \chi' + i\chi''$, derive the absorption law $I(z) = I_0 e^{-\alpha z}$ under the weak-loss approximation. Show that the same result follows from a complex refractive index $\tilde{n} = n + i\kappa$ relating each term to the real and imaginary parts of the susceptibility.

Exercise 3. Starting from the Lorentz equation of motion, derive $\chi'(\omega)$ and $\chi''(\omega)$. Determine the frequencies at which χ' changes sign and χ'' reaches its maximum in the weak-damping limit. Explain physically why the induced dipole is approximately in quadrature with the driving field on resonance.

Exercise 4. In 1902, Robert Williams Wood expanded on August Kundt's method of crossed prisms to map out the refractive index of sodium vapor near its famous "D" absorption line. In a classic crossed-prisms experiment, white light passes through a vertical glass prism, which spreads the colors out horizontally into a spectrum (like a standard rainbow). This horizontally dispersed light then passes through a horizontal prism filled with sodium vapor.





- a. Comment and explain the observations
- b. Make a model for deflection observed on the screen using Snell's law and comment how to determine the free parameters of the Lorentz model from these observations
- c. Can you observe slow light (i.e. ultra-low light velocities) in this experiment? How would you do it?

Exercise 5. In the coordinate representation and by examining whether the corresponding integrands are even or odd functions of position, use parity alone to determine whether the following electric-dipole matrix elements must vanish for a two-level atom with inversion symmetry: $\langle e|\hat{x}|e\rangle$, $\langle o|\hat{x}|o\rangle$, $\langle e|\hat{x}|o\rangle$.